

Differential Equations Lesson 01

Question 1:

$$f(x) = x^3 + 4x$$

a) $\frac{dy}{dx} = \underline{\hspace{2cm}} ?$

b) By definition of differential: $dy = \underline{\hspace{2cm}} ? \cdot dx$ and $dx = \frac{dy}{\underline{\hspace{2cm}} ?}$

c) $\frac{dy}{dx} \cdot dx = \underline{\hspace{2cm}} ?$

Question 2:

$$f(x) = \frac{5}{4}x^4 + 2x^2$$

a) $\frac{dy}{dx} = \underline{\hspace{2cm}}?$

b) By definition of differential: $dy = \underline{\hspace{2cm}}? \cdot dx$ and $dx = \frac{dy}{\underline{\hspace{2cm}}?}$

c) $\frac{dy}{dx} \cdot dx = \underline{\hspace{2cm}}?$

Question 3:

$$f(x) = \frac{3}{2}x^4 + \frac{1}{3}x^3 + 2x^2$$

a) $\frac{dy}{dx} = \underline{\hspace{2cm}}?$

b) By definition of differential: $dy = \underline{\hspace{2cm}}? \cdot dx$ and $dx = \frac{dy}{\underline{\hspace{2cm}}?}$

c) $\frac{dy}{dx} \cdot dx = \underline{\hspace{2cm}}?$

Question 4:

General Solution: $y = 4x^2 + 5x + C$

Differential Equation: $4y' - 12 = 32x + 8$.

Determine whether the general solution $y = 4x^2 + 5x + C$ satisfies the differential equation $4y' - 12 = 32x + 8$

a) $y' = \underline{\hspace{2cm}}?$

b) Does the general solution $y = 4x^2 + 5x + C$ satisfies the differential equation $4y' - 12 = 32x + 8$?

Question 5:

General Solution: $y = 4x^3 + 3x^2 + C$

Differential Equation: $\frac{1}{2}y'' + 2y' = 24x^2 + 24x + 3.$

Determine whether the general solution $y = 4x^3 + 3x^2 + C$ satisfies the differential equation $\frac{1}{2}y'' + 2y' = 24x^2 + 24x + 3.$

a) $y' = \underline{\hspace{1cm} ? \hspace{1cm}}$

b) $y'' = \underline{\hspace{1cm} ? \hspace{1cm}}$

c) Does the general solution $y = 4x^3 + 3x^2 + C$ satisfies the differential equation $\frac{1}{2}y'' + 2y' = 24x^2 + 24x + 3$?

Question 6:

Solve the differential equation $y' = \frac{dy}{dx} = 3x^2$

a) $\int 3x^2 \cdot dx = \underline{\hspace{2cm}} ?$

b) $\int \frac{dy}{dx} \cdot dx = \underline{\hspace{2cm}} ?$

c) $y = \underline{\hspace{2cm}} ?$

Question 7:

Solve the differential equation $y' = \frac{dy}{dx} = x^3 - 4x$

a) $\int (x^3 - 4x) \cdot dx = \underline{\hspace{2cm}} ?$

b) $\int \frac{dy}{dx} \cdot dx = \underline{\hspace{2cm}} ?$

c) $y = \underline{\hspace{2cm}} ?$

Question 8:

Solve the differential equation $y' = \frac{dy}{dx} = \frac{4x}{1-x^2}$

a) $\int \frac{4x}{1-x^2} \cdot dx = \underline{\hspace{2cm}} ?$

Hint: Let $u = 1 - x^2$

b) $\int \frac{dy}{dx} \cdot dx = \underline{\hspace{2cm}} ?$

c) $y = \underline{\hspace{2cm}} ?$

Question 9:

Solve the differential equation $y' = \frac{dy}{dx} = \frac{e^x}{6 - e^x}$

a) $\int \left(\frac{e^x}{6 - e^x} \right) \cdot dx = \underline{\hspace{2cm}} ?$

Hint: Let $u = 6 - e^x$

b) $\int \frac{dy}{dx} \cdot dx = \underline{\hspace{2cm}} ?$

c) $y = \underline{\hspace{2cm}} ?$

Question 10:

Solve the differential equation $y' = \frac{dy}{dx} = \cos 3x$

a) $\int \cos 3x \cdot dx = \underline{\hspace{2cm}} ?$

b) $\int 4 \cdot \frac{dy}{dx} \cdot dx = \underline{\hspace{2cm}} ?$

c) $y = \underline{\hspace{2cm}} ?$

Question 11:

Solve the differential equation $y' = \frac{dy}{dx} = 3x\sqrt{3x^2 - 5}$

a) $\int (3x\sqrt{3x^2 - 5}) \cdot dx = \underline{\hspace{2cm}} ?$ Hint: Let $u = 3x^2 - 5$

b) $\int \frac{dy}{dx} \cdot dx = \underline{\hspace{2cm}} ?$

c) $y = \underline{\hspace{2cm}} ?$

Question 12:

Solve the differential equation $y' = \frac{dy}{dx} = 3xe^{2x^2}$

a) $\int 3xe^{2x^2} \cdot dx = \underline{\hspace{2cm}}?$

Hint: Let $u = 2x^2$

b) $\int \frac{dy}{dx} \cdot dx = \underline{\hspace{2cm}}?$

c) $y = \underline{\hspace{2cm}}?$

Question 13:

Find the general solution of the differential equation $y' = \frac{dy}{dx} = \frac{1}{\sqrt{4-x^2}}$

Hint:

$$\int \frac{1}{\sqrt{a^2 - u^2}} du = \sin^{-1} \left(\frac{u}{a} \right) + C$$

Question 14:

Find the general solution of the differential equation $y' = \frac{dy}{dx} = \frac{1}{16 + x^2}$

Hint:

$$\int \frac{1}{a^2 + u^2} du = \frac{1}{a} \tan^{-1} \left(\frac{u}{a} \right) + C$$